

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble **COMPLETELY.**
12. The correct way of darkening a bubble is as :
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

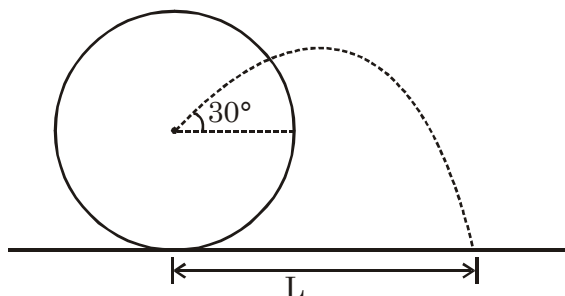
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A spherical container of radius $R = \sqrt{3}m$ is fixed on ground. 2 small holes are made at the top and at a radial position of 30° . The container is filled with water. Find the distance L (in m) at which is issuing jet strikes :

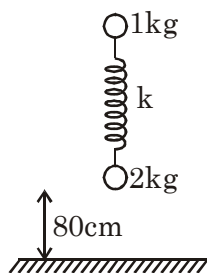


- (A) $\frac{3 + \sqrt{13}}{2}$ (B) $\frac{9 + 3\sqrt{13}}{4}$ (C) $\frac{\sqrt{13} - 1}{2}$ (D) None of these
2. The temperature of air inside the soap bubble is slightly more than the ambient temperature T , so that the bubble begins to rise? Bubble radius is r , the surface tension of the soap film σ . Weight of soap film and its thickness can be neglected. If the temperature of the air inside is $T + \Delta T$, find ΔT . Atmospheric pressure = P_0 .

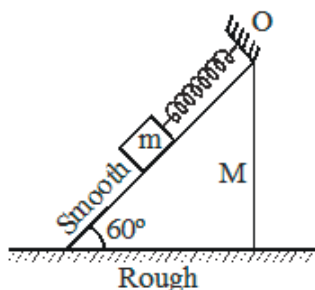
- (A) $\Delta T \geq \frac{4\sigma T}{P_0 r}$ (B) $\Delta T \geq \frac{8\sigma T}{P_0 r}$ (C) $\Delta T \geq \frac{2\sigma T}{P_0 r}$ (D) $\Delta T \geq \frac{\sigma T}{P_0 r}$

Space for Rough Work

3. A system of 2 balls and a spring are dropped from a height of 80 cm on ground as shown. The spring is initially at natural length and collision is elastic. Assuming collision to be instantaneous, what is the energy of SHM when system begins rising in air ?



- (A) $\frac{64}{3}$ J (B) 8J (C) $\frac{118}{3}$ J (D) depends on value of k
4. If $x = A \sin\left(\omega t + \frac{\pi}{6}\right)$, then find the minimum time t when kinetic energy of SHM is same as its potential energy.
- (A) $\frac{\pi}{\omega}$ (B) $\frac{2\pi}{\omega}$ (C) $\frac{\pi}{6\omega}$ (D) $\frac{\pi}{12\omega}$
5. A block of mass m is attached to a fixed position 'O' on a smooth inclined wedge of mass M . The block oscillates along the incline with amplitude A and angular frequency ω . The wedge is located on a rough horizontal surface of friction coefficient μ . Find the force of friction acting on the wedge if it always remains at rest.



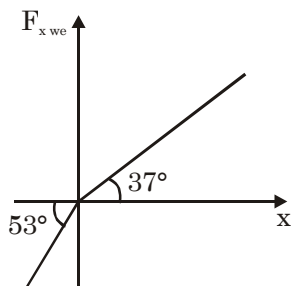
- (A) $\mu(M + m)g$ (B) $\frac{1}{2} m\omega^2 A \sin(\omega t + \phi)$
- (C) $\mu (M + m)\omega^2 A \sin (\omega t + \phi)$ (D) $\frac{1}{2} (m + M) A \omega^2 \sin (\omega t + \phi)$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. A peculiar spring has force versus extension graph as shown below. The force is measured in kN and x in dm. Negative x implies compression. This spring is connected to a block of mass 1 kg at one end and on the other end to a wall. The system is made to oscillate on a horizontal smooth surface along the spring. For this purpose, the spring is compressed by 12 cm and released.



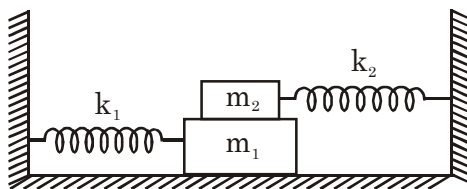
- (A) The time period of its oscillation is $\frac{7\pi}{200\sqrt{3}}$ sec.
- (B) The maximum extension is 9 cm
- (C) The maximum potential energy of the spring is 96J.
- (D) The time taken by the system to first reach the position of an extension of 8 cm is $\frac{\pi}{120\sqrt{3}}$ sec.

Space for Rough Work

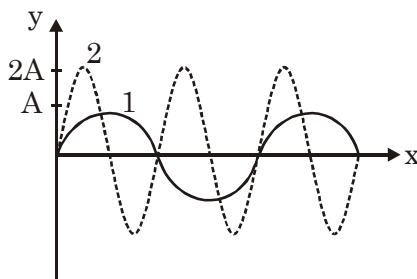
7. A plane wave $y = a \sin(bx + ct)$ is incident on a surface. Equation of the reflected wave is $y' = a' \sin(ct - bx)$ While a and a' are positive quantity, which of the following statements is **CORRECT**?
- (A) The wave is incident on the surface normally.
- (B) Reflection is taking place from y-z plane.
- (C) Medium in which incident wave is travelling is rarer than the other medium
- (D) a' cannot be greater than a .
8. A Glass capillary tube closed at one end has inner diameter of 1.5 mm and a length of 20 cm. It is dipped in vertical position in a liquid to a depth of 10 cm from the open side. Initially the capillary tube was full of air at atmospheric pressure. Contact angle $\theta = 143^\circ$. Surface tension of liquid is 0.45 J/m^2 , density of liquid is 1 gm/cc . If the final pressure of air inside is 4000 Pa .
- (A) The length of liquid column in the tube is $4/3 \text{ mm}$.
- (B) The radius of curvature of liquid surface is $\frac{15}{16} \text{ mm}$.
- (C) The length of air column is 12 cm .
- (D) The pressure of atmosphere is $\frac{11920}{3} \text{ Pa}$.

Space for Rough Work

9. Two blocks of masses m_1 and m_2 ($m_1 > m_2$), are performing SHM together with same amplitude and same time period as shown. Surface between m_1 and ground is smooth, while between m_1 and m_2 the coefficient of friction is μ : Give that $\frac{k_1}{m_1} > \frac{k_2}{m_2}$, choose **CORRECT** options :



- (A) Time period of SHM is $2\pi\sqrt{\frac{m}{k}}$, where $m = \frac{m_1 m_2}{m_1 + m_2}$ and $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$.
- (B) Time period of SHM is $2\pi\sqrt{\frac{m}{k}}$, where $m = \frac{m_1 m_2}{m_1 + m_2}$ and $k = k_1 + k_2$.
- (C) Time period of SHM is $2\pi\sqrt{\frac{m}{k}}$, where $m = m_1 + m_2$ and $k = k_1 + k_2$.
- (D) Maximum possible amplitude of this SHM is $\frac{2\mu m_2 g(m_1 + m_2)}{k_1 m_2 - k_2 m_1}$.
10. The graph shown here depicts displacement versus position graph for two progressive sine waves. The graph shown by dark line represents wave-1 and that shown by dotted line shown wave-2. Both the waves are in the same medium. Choose the **CORRECT** statement(s)



- (A) The average power carried by wave-2 is 16 times the average power carried by wave-1.
- (B) The energy contained in one wavelength of 2 is 8 times that of wave-1
- (C) The time period of wave-2 is twice that of wave-1
- (D) The maximum velocity of the particles of medium in case of wave-2 is 4 times that of wave-1

Space for Rough Work

11. A string oscillates in its third harmonic with a frequency of 195 Hz. The tension in the string is 338 N, and its length is 2 m. Which of the following options is correct?
- (A) If the string is fixed at one end only, its mass is 2.5 gm.
 (B) If the string is fixed at one end only, its mass is 5 gm.
 (C) If the string is fixed at both the ends, its mass is 10 gm.
 (D) If the string is fixed at both the ends, its mass is 20 gm.
12. A thick spherical shell of inner and outer radii r and $2r$ respectively has thermal conductivity $k = \frac{\rho}{x^n}$ where ρ is a constant and x is distance from the centre of the shell. The inner and outer walls are maintained at temperature T_1 and $T_2 (< T_1)$.
- (A) The value of number n for which the temperature gradient remains constant throughout the thickness of the shell is $n = 2$.
 (B) For this value of n , the distance x at which the temperature is the average of inner and outer shell is $3r/2$.
 (C) For this value of n , the rate of heat flow is $2\pi\rho \frac{(T_1 - T_2)}{r}$.
 (D) For this value of n the temperature gradient is $\frac{(T_2 - T_1)}{r}$.
13. A thin uniform rod of mass M is moving in a plane and has a kinetic energy of $\frac{4}{3} MV^2$ where V is the speed of its centre of mass. The speed of any of its point can be :
- (A) 0.5 V (B) 1.5 V (C) 2.5 V (D) 3.5 V

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

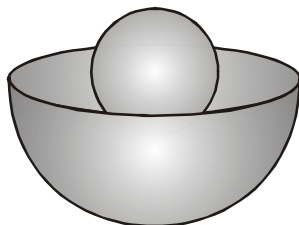
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
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9	9	9	9	9	9	9	9	9	9

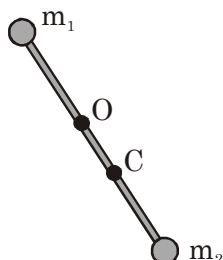
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A spherical blackbody of radius R is arranged concentrically inside a hemispherical blackbody of radius $2R$ as shown in the figure. If the hemispherical body is maintained at a temperature T_0 , and the steady state temperature of the sphere is T . Assume that the sphere is perfectly conducting and the only mode of heat transfer between the sphere and the rest of the universe is radiation. The temperature of the enclosure around the system is assumed to be zero Kelvin. Fill value of T_0/T .



Space for Rough Work

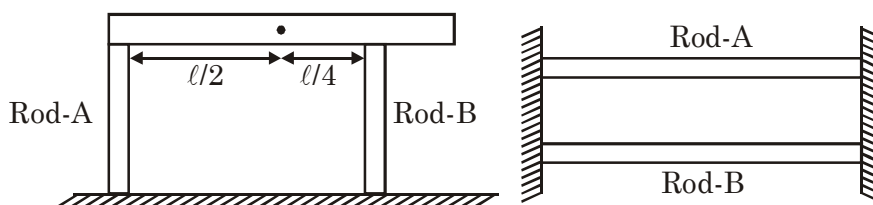
2. Dumbbell of length ℓ with masses m_1 and m_2 at the ends being suspended at point O, makes small vibrations about its stable equilibrium position while remaining in its plane passing through the suspension point (figure) Find the frequency (in Hz) of these vibrations, if the suspension point is at a distance d from the centre of mass C. ($m_1 = 3\text{gm}$, $m_2 = 5\text{gm}$, $\ell = 80\text{ cm}$, $d = 10\text{ cm}$.)



3. A long string of mass per unit length 0.2 Kg/m is stretched to a tension 500N . A transverse wave of amplitude $A = 10\text{ mm}$ and $\lambda = 0.5\text{ m}$ is traveling on this string. This string is joined to another string of same cross section and mass per unit length 0.8 Kg/m . The fraction of the average power carried by the wave in the first string that is transmitted to the second string is K . Find K .

Space for Rough Work

4. A solid cylinder of height 15 cm and density $2000/3 \text{ kg/m}^3$ is held vertically in air above the surface of a lake such that its lower flat surface is touching the surface of water. It is now released. After what minimum time (in sec) will it come to rest again. Assume that it always remains vertical and neglect viscosity or any other energy losses.
5. A uniform heavy plank of length L is resting on two rods of identical length and area of cross section, made up of different materials. It is observed that to keep the plank horizontal, one rod is to be at the end of the plank (say rod A) and another rod (say rod B) is to be at separation $L/4$ from centre of plank as shown in figure. Now the rods are kept between two rigid walls such that separation between the walls is equal to the natural length of the rod. Now temperature of both the rods is gradually increased in same manner, it is observed that thermal stress developed in rod A is always thrice the thermal stress developed in rod B. Ratio of coefficient of linear expansion of rod A to coefficient of linear expansion of rod B will be :



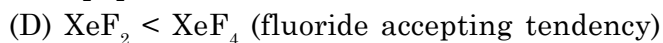
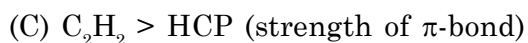
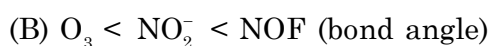
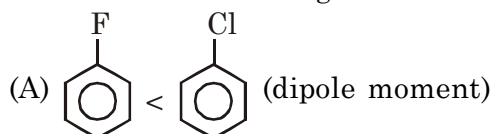
Space for Rough Work

PART-2 : CHEMISTRY

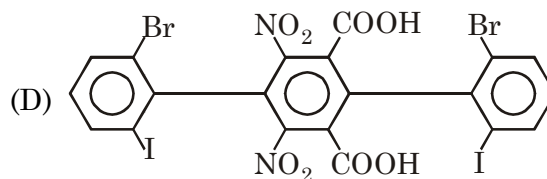
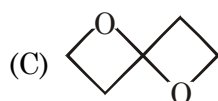
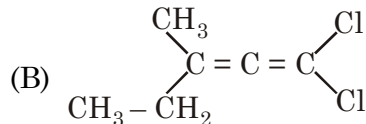
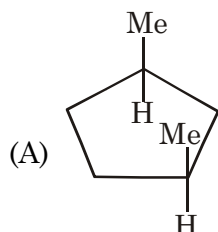
SECTION-I(i) : (Maximum Marks : 15)

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1. Which of the following is **INCORRECT**?

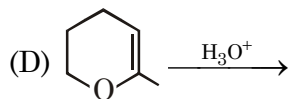
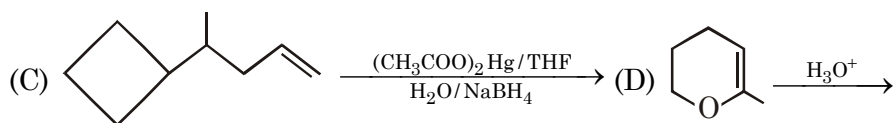
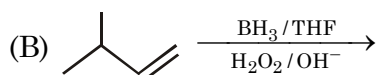
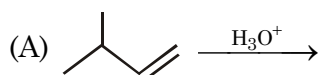


2. Which of the following molecule is resolvable?



Space for Rough Work

3. Which of following gives 2° alcohol major product?



4. The density of a gaseous substance at 1 atm pressure and 773K is 0.4 g/l. If the molecular mass of the substance is 30, then dominant forces existing among gas molecules in this condition is -

(A) Repulsive (B) Attractive (C) Both (D) None

5. Total number of electrons present in 96 amu of SO_4^{2-} ion -

(A) $50N_A$ (B) 50 (C) $\frac{50}{N_A}$ (D) 100

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
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6. Which of the following statements(s) is/are correct for a vanderwaal gas ?

(A) $Z = \frac{V_m}{V_m - b} - \frac{a}{V_m RT}$

(B) $Z = 1.0$ at all pressures at Boyle temperature

(C) $Z = 3/8$ at all pressures at critical temperature

(D) Below Boyle temperature, the gas may behave as super critical fluid

7. Identify the correct statement(s)

(A) The e/m ratio of canal rays particle is same irrespective of the gas taken

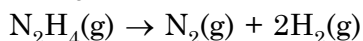
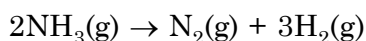
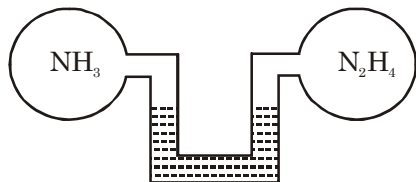
(B) Cathode rays are deflected in both electric and magnetic fields

(C) Anode rays are produced as a result of ionisation of the gas in discharge tube

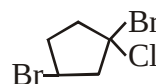
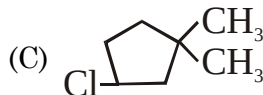
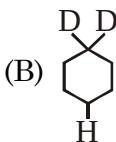
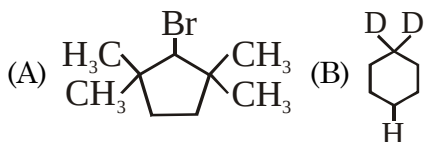
(D) The e/m ratio of cathode rays particles is same irrespective of the gas taken.

Space for Rough Work

8. Figure shows a manometer, one arm is connected with a bulb containing NH_3 and other arm containing N_2H_4 , both at initial pressure of 1 atm. Initially both arm of manometer contain liquid at same level. What will be the difference in level of arms, when 50% $\text{NH}_3(\text{g})$ and 75% of $\text{N}_2\text{H}_4(\text{g})$ dissociates according to the given reaction. Assume temperature remains constant.



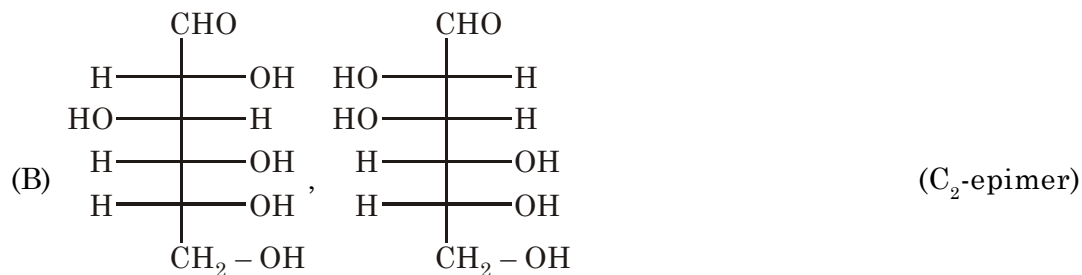
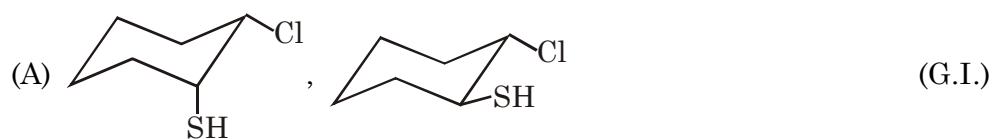
- (A) 76 cm if liquid having density 13.6 gm/ml is used
 (B) 38 cm if liquid having density 27.2 gm/ml is used
 (C) 152 cm if liquid having density 6.8 gm/ml is used
 (D) 304 cm if liquid having density 3.4 gm/ml is used
9. Which of the following compounds do not have any geometrical isomers.



10. Which of the following is / are **incorrect** IUPAC names
- (A) 2-Ethyl-3-methyl pentane (B) 1-amino butane-1-one
 (C) 2-Methyl-3-ethyl pentane (D) 4-(1, 1-dimethyl ethyl)-2-methyl pentane

Space for Rough Work

11. Which of following relationship between given isomeric pair correctly given :



Space for Rough Work

12. For a real gas following equation -

$$\left(P + \frac{\alpha}{TV_m^2}\right)(V_m - \beta) = RT; \text{ where } \alpha \text{ \& } \beta \text{ are positive gas constant, identify the correct}$$

statements(s) -

(A) The temperature above which the gas cannot be liquified by application of pressure

is $\sqrt{\frac{8\alpha}{27R\beta}}$

(B) The minimum pressure needed for liquification of gas at temperature calculated above is

$\sqrt{\frac{\alpha R}{216\beta^3}}$

(C) The molar volume of the gas at the temperature & pressure calculated above is 3β

(D) The molar volume of the gas at the temperature & pressure calculated above is 9β

13. An orbital "A" has a nodal plane which is common with another orbital "B". The nodal plane

is PQ ($P, Q \in \{x, y, z\}$). The orbital A belongs to a subshell that serves as second excited state

for $H^\ominus$. ($x, y, z \longrightarrow$ cartesian axes)

The orbital B is of double dumbbell shape.

Which of the following is/are **CORRECT**.

(A) A & B can never form a delta bond

(B) If A is oriented in "y" direction and B has zero electron density in "y" direction, then A & B can never form pi bond

(C) If A is oriented in 'x' direction & B has zero electron density in 'x' direction then A & B can form bonds in exactly 2 conditions

(D) "A" can form a bond with 's' orbital & "B" can never form bond with 's' orbital along any axis P or Q.

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

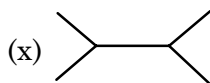
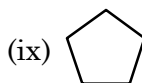
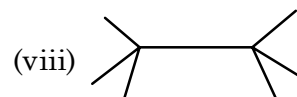
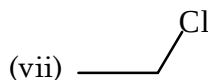
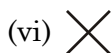
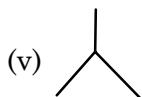
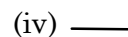
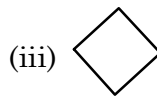
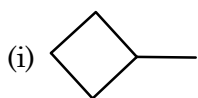
- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
●	●	0	0	0	●	●	0	0	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	●	2	2	2	2	●
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	●	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

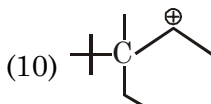
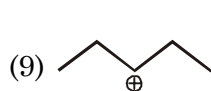
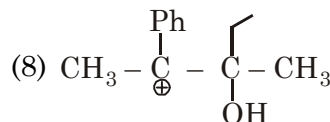
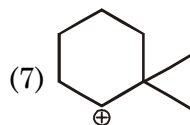
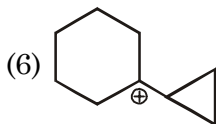
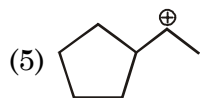
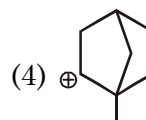
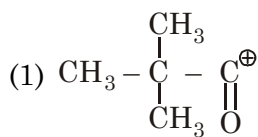
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Number of specie(s) among the following which do not have X-O-O-X linkage.
Thiosulphuric acid, Persulphuric acid, Chlorine heptoxide, Dioxygendifluoride, Sodium perdisphosphate, Sodium peroxyborate, Borax, Perphosphoric acid.
- Total number of sp^3 hybrid orbitals in γ -SO₃ is _____.
- Number of compound which form single monochloroderivatives on monochlorination.

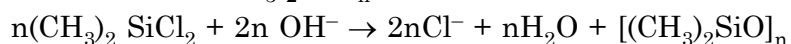


Space for Rough Work

4. Which of following carbocation shows rearrangement?



5. In a process of water - proofing, a fabric is exposed to $(\text{CH}_3)_2\text{SiCl}_2$ vapour. The vapour reacts with hydroxyl groups on the surface of the fabric or with traces of water to form the water proofing film $[(\text{CH}_3)_2\text{SiO}]_n$ by the reaction.



where n is a large integer. The water proofing film is deposited on the fabric, layer upon layer. Each layer is $6\text{\AA}$ thick (the thickness of $[(\text{CH}_3)_2\text{SiO}]$). How much $(\text{CH}_3)_2\text{SiCl}_2$ (in gm) is needed to water proof one side of a piece of fabric, $1\text{m} \times 3\text{m}$, with a film 296 layers thick. The density of film is 1 gm/cm^3 [Given : $\text{Si} = 28$, $\text{Cl} = 35.5$]

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the roots of the equation $(a + b + c)x^2 - (b + 2c)x + c = 0$ are

(A) $\frac{\alpha+1}{\alpha}, \frac{\beta+1}{\beta}$

(B) $\frac{\alpha-1}{\alpha}, \frac{\beta-1}{\beta}$

(C) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

(D) $\frac{\alpha}{\alpha-1}, \frac{\beta}{\beta-1}$

2. If sum $S = \sum_{r=1}^{2020} \frac{(-1)^{r+1}}{r}$, then S is also equal to -

(A) $\sum_{r=1011}^{2020} \frac{(-1)^r}{r}$

(B) $\sum_{r=1011}^{2020} \frac{1}{r}$

(C) $\sum_{r=1011}^{2020} \frac{(-1)^{r+1}}{r}$

(D) $\sum_{r=1010}^{2020} \frac{1}{r}$

Space for Rough Work

3. Locus of a point which moves such that tangents drawn from it to $y^2 = 4x + 4$ and $y^2 = 8x + 16$ are mutually perpendicular, is-
- (A) $y = x + 2$ (B) $y = x + 3$ (C) $x + 3 = 0$ (D) $x + 2 = 0$
4. The eccentricity of the ellipse $\left(\frac{x-3}{y}\right)^2 + \left(1 - \frac{4}{y}\right)^2 = \frac{1}{9}$, is
- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{9}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$
5. If in $\triangle ABC$; $AB = 4$, $BC = 6$ and $AC = 5$; h_1, h_2, h_3 be the length of altitude of from vertices A, B, C respectively, then value of $\left(\frac{1}{h_1} + \frac{1}{h_2} - \frac{1}{h_3}\right)$ is equal to-
- (A) $\frac{\sqrt{7}}{15}$ (B) $\frac{2\sqrt{7}}{15}$ (C) $\frac{4\sqrt{7}}{15}$ (D) $\frac{8\sqrt{7}}{15}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. If $(x-1)^2 + (y-1)^2 = 1$, then $\frac{y}{x+\sqrt{3}-1}$ can be-
- (A) 0 (B) 1 (C) 2 (D) 3
7. In a triangle ABC if $4\sin A = 4\sin B = 3\sin C$ and perimeter of triangle is 10, then
- (A) area of triangle is $2\sqrt{5}$ (B) in radius of triangle is $\frac{2}{\sqrt{5}}$
- (C) circumradius of triangle is $\frac{2\sqrt{5}}{9}$ (D) smallest altitude of triangle is $\sqrt{5}$

Space for Rough Work

8. Let $x + y = 2$ and $2x + y = 4$ are tangents to a parabola such that $P(-2, 4)$ and $Q(1, 2)$ are foot of perpendicular drawn from focus to these tangents respectively, then -
- (A) equation of axis of parabola is $3x - 2y + 21 = 0$
 (B) equation of latus rectum is $2x + 3y + 27 = 0$
 (C) length of latus rectum = $\frac{65}{\sqrt{13}}$
 (D) coordinates of focus is $(-9, 3)$
9. The equation of a circle(s) in which the chord joining the points $(1, 2)$ and $(2, -1)$ subtends an angle of $\frac{\pi}{4}$ at any point on the circumference is-
- (A) $x^2 + y^2 - 5 = 0$ (B) $x^2 + y^2 - 6x - 2y + 5 = 0$
 (C) $x^2 + y^2 + 6x + 2y - 15 = 0$ (D) $x^2 + y^2 + 7x - 2y + 14 = 0$
10. Which of the following determinant(s) reduces to zero-

(A) $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$

(B) $\begin{vmatrix} 1 & 2 & 3 \\ 51 & 52 & 53 \\ 201 & 202 & 203 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & b-c \\ b-a & 0 & c-a \\ c-b & a-c & 0 \end{vmatrix}$

(D) $\begin{vmatrix} a & \lambda a + \mu b & b \\ p & \lambda p + \mu q & q \\ u & \lambda u + \mu \omega & \omega \end{vmatrix}$

Space for Rough Work

11. A hyperbola intersects an ellipse $x^2 + 9y^2 = 9$ orthogonally. The eccentricity of the hyperbola is reciprocal of that of ellipse. If the axes of the hyperbola are along coordinate axes, then-
- (A) vertices of hyperbola are $\left(\pm \frac{8}{3}, 0\right)$
- (B) y coordinate of point of intersection of ellipse and hyperbola is either $\frac{1}{3}$ or $-\frac{1}{3}$
- (C) latus rectum of hyperbola is $\frac{2}{3}$
- (D) latus rectum of hyperbola is $\frac{4}{3}$
12. Which of the following hold(s) good ?
- (A) $(\log_3 5)(\log_3 4)(\log_{10} 11) > 1$
- (B) $(\log_2 3)(\log_{1/3} 5)(\log_4 6) < 1$
- (C) $\log_{100} 7 > \log_{29} 3$
- (D) $\log_{\pi} 5 > \log_e 5$
13. If $f(n, \theta) = \sin \theta \sin^2 \frac{\theta}{2} + \frac{\sin 2\theta \sin^2 \theta}{2} + \frac{\sin 4\theta \sin^2 2\theta}{4} + \dots + \frac{\sin(2^n \theta) \sin^2(2^{(n-1)} \theta)}{2^n}$; $n \in \mathbb{N} \cup \{0\}$, then
- (A) $f\left(5, \frac{\pi}{6}\right) = \frac{64 - \sqrt{3}}{128}$
- (B) $f\left(5, \frac{\pi}{6}\right) = \frac{64 - \sqrt{3}}{256}$
- (C) $f\left(2, \frac{\pi}{8}\right) = \frac{\sin(\pi/8)}{2}$
- (D) $f\left(2, \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If two lines, passing through (4,3), form an equilateral triangle with the line $3x - 4y + \sqrt{15}\sqrt{3} = 0$, then area of this triangle is
2. Length of the normal chord of the parabola $y^2 = 4x$, which makes an angle $\frac{\pi}{4}$ with x-axis is-(take $\sqrt{2} = 1.41$)

Space for Rough Work

3. For all admissible values of the parameter 'a' the straight line $2ax + y\sqrt{1-a^2} = 1$ will always touch an ellipse whose principal axes are along coordinate axes and having eccentricity is equal to-
4. Common tangents of $9x^2 - 9y^2 = 8$ and $y^2 = 32x$ passes through (a,0), then a is-
5. The number of solution(s) of trigonometric equation $4\cos^2x + \tan^2x + \cot^2x + \sec^2x = 6$ in $[0, 2\pi]$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	5	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	15
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator